

A Game-Native Model of the Gold–Heisenberg Extension

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Abstract

The extraspecial dictionary identifies quadratic maps Q on an $\mathbb{F}_2$ -space with central extensions whose squares read Q and whose commutators read its polar form. The remaining structural question for the Gold trace family asked for such an extension built from game values and game operations, without importing the diagonal $Q_a(e_i)$ or a unary Q_a oracle. We give a basis-free answer. The oriented half-polar term

$$\phi_{\lambda,a}(x, y) = \text{Tr}(\lambda xy^{2^a})$$

is a bilinear two-cocycle. Its cocycle group on $\mathbb{F}_2 \times \mathbb{F}_{2^m}$ is represented by two nimber-game registers; the product uses only disjunctive sum, Turning-Corners nim multiplication, Frobenius, and trace. Its squares are the scaled Gold form and its commutators are the Gold polar form. A signed-translation action on finite bundles of short-game values makes the composition literal and faithful. We also correct the target’s terminology: an unscaled Gold form has a polar radical, so its full extension is extraspecial-type, not extraspecial. In the power-of-two nimber tower the radical has a canonical central lift; quotienting by it yields the genuine extraspecial core, with the first $(m, a) = (4, 1)$ cell equal to Q_8 . The universal cocycle algebra is kernel-checked in Lean 4.

1 The target and the oriented half

Let $K = \mathbb{F}_{2^m}$ be represented by its normal-play nimber values. Fix $a \geq 0$ and $\lambda \in K$. The scaled Gold trace form is

$$Q_{\lambda,a}(x) = \text{Tr}_{K/\mathbb{F}_2}(\lambda x^{1+2^a}).$$

Its polar form is

$$B_{\lambda,a}(x, y) = Q_{\lambda,a}(x + y) + Q_{\lambda,a}(x) + Q_{\lambda,a}(y).$$

The structural extension problem is not solved by the symmetric form B alone: characteristic two keeps the diagonal independent. The missing datum is already present before symmetrization.

Definition 1 (Gold half-polar cocycle). Define

$$\phi_{\lambda,a}(x, y) = \text{Tr}_{K/\mathbb{F}_2}(\lambda xy^{2^a}).$$

Then

$$\begin{aligned} \phi_{\lambda,a}(x, x) &= Q_{\lambda,a}(x), \\ \phi_{\lambda,a}(x, y) + \phi_{\lambda,a}(y, x) &= B_{\lambda,a}(x, y). \end{aligned}$$

Frobenius is additive and trace is linear, so ϕ is biadditive. This one observation supplies both the construction and its associativity.

2 The game-native extension

Theorem 2 (Gold–Heisenberg game model). **[PROVED]** *On*

$$\mathcal{E}_{\lambda,a} = \mathbb{F}_2 \times K$$

put

$$(s, x) \diamond (t, y) = (s + t + \phi_{\lambda,a}(x, y), x + y). \quad (1)$$

Let $z = (1, 0)$ and $\pi(s, x) = x$. Then:

(i) $\mathcal{E}_{\lambda,a}$ is a group, and

$$1 \longrightarrow \langle z \rangle \longrightarrow \mathcal{E}_{\lambda,a} \xrightarrow{\pi} K^+ \longrightarrow 1$$

is a central extension with two-point fibers. The identity is $(0, 0)$ and

$$(s, x)^{-1} = (s + Q_{\lambda,a}(x), x).$$

(ii) Its squares and commutators are

$$(s, x)^2 = z^{Q_{\lambda,a}(x)}, \quad (2)$$

$$[(s, x), (t, y)] = z^{B_{\lambda,a}(x,y)}. \quad (3)$$

Thus $\mathcal{E}_{\lambda,a}$ is exactly the extension $E_{Q_{\lambda,a}}$; every coordinate value $Q_{\lambda,a}(e_i)$ is produced as the square of the lift $(0, e_i)$.

(iii) The product is one fixed game construction. Represent s by the impartial value 0 or $$, and x by the nimber game $*x$. In these two registers,¹*

$$(C, X) \diamond (D, Y) = \left(C \oplus D \oplus \text{Tr}_{\text{game}} \left((*\lambda) \underset{\text{nim}}{\otimes} X \underset{\text{nim}}{\otimes} \text{Frob}^a(Y) \right), X \oplus Y \right). \quad (4)$$

Here $\oplus$ is disjunctive sum of impartial values, $\underset{\text{nim}}{\otimes}$ is Conway's Turning-Corners product, Frobenius is repeated diagonal Turning Corners, and trace is the disjunctive sum of the Frobenius iterates. The formula contains no diagonal table, unary $Q_{\lambda,a}$ call, outcome test, or input-dependent branch.

(iv) The center is

$$Z(\mathcal{E}_{\lambda,a}) = \{(s, r) : s \in \mathbb{F}_2, r \in \text{rad } B_{\lambda,a}\}.$$

Consequently the group is extraspecial exactly when the polar form is nondegenerate and has positive rank. Every positive-rank nonsingular complement W of the radical gives an extraspecial subgroup $\mathbb{F}_2 \times W$ of order $2^{1+\dim W}$, with squaring form $Q_{\lambda,a}|_W$.

Proof. Biadditivity gives the normalized cocycle identity

$$\phi(x, y) + \phi(x + y, w) = \phi(y, w) + \phi(x, y + w).$$

The two bracketings of three factors in (1) have the same vector coordinate, and this identity makes their central coordinates equal. Thus the product is associative. The asserted identity and inverse

¹The first register is restricted to the prime subfield $\{0, *\}$; the trace expression lands there structurally.

follow directly because both additive coordinates have exponent two. The element z is central of order two, π is onto, and each fiber is the pair (s, x) , $z \diamond (s, x)$.

Putting the same input in both slots gives

$$(s, x)^2 = (\phi(x, x), 0) = (Q_{\lambda, a}(x), 0),$$

which proves (2). Swapping the order of two factors changes only the central coordinate, by $\phi(x, y) + \phi(y, x) = B_{\lambda, a}(x, y)$, proving (3). The standard central-extension dictionary now identifies the group with $E_{Q_{\lambda, a}}$.

Every operation in (4) is the displayed game realization of the corresponding field operation, so the same calculation proves the game-register statement. Finally, (s, x) commutes with every (t, y) exactly when $B(x, y) = 0$ for every y , which is the center formula. On a positive-rank nonsingular complement the center is $\langle z \rangle$, a nonzero commutator generates it, and the finite 2-group identity $\Phi(G) = G^2[G, G]$ gives $Z(G) = [G, G] = \Phi(G) = \langle z \rangle$. $\square$

The fixed binary half ϕ is essential. Squaring a lift necessarily reveals Q in every model of E_Q , so “without a Q evaluator” cannot coherently forbid multiplication from computing the diagonal of its own product. The nonvacuous condition is the one Theorem 2 meets: multiplication must not import Q , its basis diagonal, or an outcome-labelled lookup table.

3 A literal composition on game-valued contexts

The registered product already answers the stated question. There is also a faithful operator model showing that its multiplication is actual composition of game-built contexts rather than an abstract group law with game labels attached.

Let $\mathcal{G}$ be the additive group of short partizan game values and $\mathcal{C} = \mathcal{G}^K$, the finite bundle of game values indexed by the nimber field. Define

$$(zF)(u) = -F(u), \quad (T_x F)(u) = (-1)^{\phi_{\lambda, a}(u, x)} F(u + x). \quad (5)$$

The coordinate translation uses nim addition; the sign is ordinary game negation.

Proposition 3 (faithful signed translations). **[PROVED]** *With composition ordered as $T_x T_y = T_x \circ T_y$,*

$$T_x T_y = z^{\phi_{\lambda, a}(x, y)} T_{x+y}.$$

Hence $(s, x) \mapsto z^s T_x$ is a faithful homomorphism

$$\mathcal{E}_{\lambda, a} \hookrightarrow \text{Aut}(\mathcal{C}).$$

Proof. Biadditivity gives

$$\begin{aligned} (T_x T_y F)(u) &= (-1)^{\phi(u, x) + \phi(u+x, y)} F(u + x + y) \\ &= (-1)^{\phi(u, x+y) + \phi(x, y)} F(u + x + y), \end{aligned}$$

which is the multiplication law. For faithfulness, let δ_0 take the integer-game value 1 at 0 and 0 elsewhere. The support of $z^s T_x \delta_0$ is the singleton $\{x\}$, and its value there has sign $(-1)^{s+Q_{\lambda, a}(x)}$. Thus the image determines both x and s . $\square$

This is a representation of the structural multiplication, not the forbidden Cayley move rule $g \rightarrow gn$ with fixed n : play still uses the flagship paper’s automorphism-equivariant squaring rule once the extension is built.

4 Radical discipline and the genuine extraspecial core

The phrase “extraspecial E_{Q_a} ” hid a mathematical defect. The full unscaled Gold polar form is always singular. The central extension above is canonical on the full field; extraspeciality belongs to its nonsingular core.

Proposition 4 (Gold radical). **[STANDARD MATH]** *For the unscaled form Q_a on $K = \mathbb{F}_{2^m}$, put $d = \gcd(2a, m)$. Then*

$$\text{rad } B_a = \mathbb{F}_{2^d}, \quad \text{rank } B_a = m - d.$$

In particular every unscaled Gold extension has center larger than $\langle z \rangle$; when $m = 2^k$ and $a \geq 1$, it is never itself extraspecial.

Proof. Frobenius invariance and trace adjointness give, with exponents read modulo m ,

$$B_a(x, y) = \text{Tr}\left((x^{2^{m-a}} + x^{2^a})y\right).$$

The trace pairing is nondegenerate. Hence x is radical exactly when $x^{2^{m-a}} = x^{2^a}$, equivalently $x^{2^{2a}} = x$. The fixed field is $\mathbb{F}_{2^{\gcd(2a, m)}}$. $\square$

For a general degree, if Q_a is nonzero on the radical then no quadratic form descends to the quotient; one must choose a nonsingular complement, and its Arf type may depend on that choice. The power-of-two tower used by the project lies in the cleaner case.

Theorem 5 (canonical power-of-two radical quotient). **[PROVED]** *Assume m is a power of two, $a \geq 1$, and $d = \gcd(2a, m) < m$. Put $R = \mathbb{F}_{2^d}$ and*

$$H = \{(0, r) : r \in R\} \subseteq \mathcal{E}_a.$$

Then $Q_a|_R = 0$ and $\phi_a|_{R \times R} = 0$. The subgroup H is central and elementary abelian, and

$$1 \longrightarrow \langle \bar{z} \rangle \longrightarrow \mathcal{E}_a/H \longrightarrow K/R \longrightarrow 1$$

is a genuine extraspecial extension of order 2^{1+m-d} with the induced nonsingular Gold quadratic form. For $(m, a) = (4, 1)$ it is Q_8 .

Proof. The relative degree m/d is even. For $r, s \in R$, both rs^{2^a} and r^{1+2^a} lie in R , while absolute trace from K restricts on R to (m/d) times the absolute trace of R . It therefore vanishes, proving both restrictions. Proposition 4 makes H central, and $\phi|_{R \times R} = 0$ makes it a subgroup. The quotient quadratic form is well defined because $Q|_R = B(R, K) = 0$; its polar form has zero radical by construction. Theorem 2 therefore makes the quotient extraspecial and gives its order. At $(4, 1)$ the rank is two and the already established Gold core has Arf invariant one, so the order-eight group is the minus cell Q_8 . $\square$

Thus the proposed Q_8 ordinal-sum search is unnecessary: the quaternion cell is the first canonical radical quotient of the trace-cocycle model. On any anisotropic plane more directly, lifts i, j satisfy $i^2 = j^2 = z$ and $ij = zji$, which is the standard presentation of Q_8 ; hyperbolic planes give D_8 .

5 Exact boundary and formal status

Remark 6 (naturality boundary). **[BOUNDARY]** For the unscaled form, the construction is basis-free and Frobenius covariant:

$$\phi_a(x^{2^j}, y^{2^j}) = \phi_a(x, y).$$

It is not ambient-coherent under even-degree subfield inclusions, because absolute trace restricts to zero there. This agrees with, rather than evades, the flagship paper’s divisibility no-go for one coefficient-faithful datum on the full additive group of short games. Nor can a lift section be fixed by every extension automorphism: central twists exchange its two lifts. Neither stronger condition belonged to the model problem.

Theorem 2 satisfies the stated target literally: its positions are pairs of game values, its multiplication is a fixed composition of named game constructions, and its diagonal is produced by squaring. A stronger demand that the carrier be one untagged Conway value under one pre-existing binary compound would be a different problem. It would also require a precise congruence: ordinal sum does not descend to ordinary game values, while arbitrary quotients of literal game forms can import any finite presentation and are therefore too permissive.

The universal algebra is kernel-checked in `formal/Ogdoad/GoldExtraspecial.lean`. For an arbitrary biadditive $\phi : V \rightarrow V \rightarrow \mathbb{F}_2$, Lean constructs the group, proves cocycle associativity, central retagging, the diagonal square identity, the symmetrized swap/commutator defect, the exact center criterion, noncommutativity from nonzero polar value, and order-four lifts from a diagonal-one input. It does not encode the concrete `u128` nimber field, Turning-Corners implementation, or finite-field trace; those are the paper-level specialization and the independently tested Rust surface.

References

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